
Proposed Solution Architecture

1. Solution Overview

ShopEEZ is designed as a single-page application (SPA) client connected to a secure REST API server. The application is built entirely using the MERN stack, offering a scalable, fully JavaScript-driven codebase from database models to client interfaces.

2. Route Layout & Navigation

The application features structured, secure React Router routes:

- Public Catalog Routes: Homepage (/) and Product Details (/product/:id) with search and filters.
- Customer Workspaces (Private): Cart (/cart), Wishlist (/wishlist), Checkout (/checkout), Account Profile (/profile), and Orders (/orders).
- Admin Console (Private/isAdmin): Dashboard (/admin), Inventory (/admin/products), and Order Updates (/admin/orders).

3. Core Operations & Design Decisions

Key design choices include local multi-image uploads (Multer) saving files directly to local server storage, and Redux cart serialization to LocalStorage, ensuring full offline cart persistence without hitting backend tables on every item adjustment.